Homelab architecture

Physical hosts, service placement and the shared platform.

People & devices
Browser · phone · workstation
Home LAN or remote Tailscale

Two access paths
Public: Internet → router → Caddy → selected services
Private: LAN / Tailscale → Caddy or native LAN protocols
Tailscale subnet routing runs on all three Proxmox hosts.

Cloudflare
DNS-only records · DNS-01 TLS
Private defaults; public exceptions
DNS is not a traffic tunnel.

slate · external GCP VM
Caddy · Uptime Kuma · ntfy
Independent monitoring & alerts
Checks home over Tailscale

Home router / LAN fabric — Ethernet: 1 Gb/s full duplex verified at px10, px20 and px30 — Wi-Fi → bulb & AC

px10 — Dell OptiPlex 5060

i5-8400T / 6 cores · 32 GB RAM · ZFS OS + data pools

- **dev · VM 100**
  Codex · Claude Code · Hermes · OpenCode
- **observability · LXC 104**
  Beszel · Grafana · VictoriaMetrics
- **store · LXC 109**
  Samba: AV · BACKUP · ISO
- **dns · LXC 102**
  AdGuard Home → Unbound recursion
- **auth · LXC 103**
  Pocket ID / OIDC · Tinyauth / forward-auth
- **s3 · LXC 105**
  Cairn · RustFS / S3-compatible storage
- **haos · VM 107**
  Home Assistant · Mosquitto · local device control
- **k8s-201 · VM 201**
  Talos · control plane + workloads

px20 — Dell OptiPlex 5060

i5-8400T / 6 cores · 32 GB RAM · ZFS OS + data pools

- **ctr · VM 204 / Docker + Portainer**
  Media Jellyfin · qBittorrent · Motrix
  Personal Immich · Karakeep · OpenCloud · Paperless
  Notes Nivra · separate friend instance
  Build Forgejo · Docker Registry · mirror worker
  Automate n8n · SearXNG · isolated sandbox
  Agents OpenViking · Ollama · Omni
- **orva · VM 106**
  Serverless functions · Quick Share
- **proxy · LXC 101**
  Caddy · HTTPS routing / TLS
- **k8s-202 · VM 202**
  Talos · control plane + workloads
  4 vCPU · up to 16 GiB RAM · dedicated 500 GiB data disk

px30 · maintenance — Dell OptiPlex 7040

i5-6500TE / 4 cores · 32 GB RAM · ZFS OS + data pools

- **k8s-203 · VM 203 · stopped**
  Talos · control plane + workloads
  4 vCPU · up to 16 GiB RAM · dedicated 500 GiB data disk

Placement is a snapshot; managed guests can relocate.

Kubernetes: two active nodes; k8s-203 stopped for hardware swap

Talos management: Omni on ctr. Kubernetes desired state: GitHub → Argo CD.

Networking & operations

Cilium CNI · L2 LoadBalancer services · CoreDNS
Argo CD · Headlamp · Homepage · metrics-server
vmagent · kube-state-metrics · node metrics

Persistent storage

Longhorn: both active disks on k8s-201 / k8s-202
Current volumes: 2 storage replicas each
Data crosses hosts; replication is not backup.

Supabase

Configured: 1 primary + 2 standbys; swap: 1 standby pending
Studio · Auth · REST · Realtime · Edge Functions
Storage / imgproxy · Kong · pooler · logs / analytics

Storage boundaries

ZFS → VM/LXC disks and selected 5-minute replicas
USB on px10: 4 TB AV / BACKUP + 256 GB ISO
ctr local data → app state; S3 → Cairn / RustFS objects

Availability boundaries

Proxmox HA manages the listed non-Kubernetes guests.
USB shares need physical-disk recovery after host loss.
DNS, Caddy and several apps remain single instances.

Smart home

Home Assistant → Wipro bulb / Tuya LAN
Home Assistant → Samsung AC / local DTLS
Frigate → camera feeds; HA → viewing / automation

Logical LAN fabric shown; router, switch/AP models and exact cabling have not been audited.

How traffic and data move

The important connections, without drawing every packet or replica.

Web access & identity

Client
LAN / Tailscale for private apps
Internet for selected public apps

→

Caddy · proxy LXC
HTTPS termination → backend routes
Source restrictions per service

→

Application
Native login / API credentials
No blanket public or SSO bypass

Pocket ID + Tinyauth · auth LXC
Native OIDC where supported
Caddy forward-auth for selected browser routes

Name lookup: AdGuard → Unbound

Identity is a policy dependency; API/MCP routes retain their own authentication.

GitOps & cluster operations

GitHub · Homelab repository
Declarative non-secret configuration
Reviewed changes → desired state

⇢

Argo CD · Kubernetes
Reconciles tracked applications
Signed public webhook; private UI

⇢

Talos / Kubernetes
Cilium network · Longhorn volumes
k8s-201 / 202 active; k8s-203 stopped

⇠

Omni · ctr
Manages the three Talos machines
Runs on ctr, not inside Kubernetes

Longhorn: storage-level replicas. CloudNativePG: separate PostgreSQL streaming replicas and failover.

Agents & shared memory

dev · four agents
Codex · Claude Code
Hermes · OpenCode

→

OpenViking · ctr
Shared account-scoped memory
Native plugins / provider and MCP

→

Ollama · ctr
Local embedding service
Not the agents' primary chat model

Tools & integrations
Orva · Nivra · Supabase · browser / developer MCPs
Hermes → Karakeep CLI / Home Assistant API

Hermes: Go / DeepSeek + Codex fallback. OpenCode: Claude Agent SDK. Telegram / iMessage → Hermes.

Media & object storage

Media path
qBittorrent / Motrix → Store AV share → Jellyfin
Encrypted SMB3 directly over the LAN; not through Caddy.
Media stays on Store's USB SSD, not ctr's OS disk.

Temporary sharing
Quick Share CLI → Orva function → Cairn S3
Short Orva share URL → function proxy → object download.
Cairn is a separate native service on the s3 LXC.

Container images
Docker client → Registry → RustFS S3
Registry runs on ctr; image layers live in RustFS.
Cairn and RustFS share one LXC / host failure domain.

Proxmox nodes use Store BACKUP / ISO over SMB directly. App databases remain on their own local or block volumes.

Home automation & visibility

Hermes / user
Home Assistant API / app
Existing authenticated access

→

Home Assistant · haos
Wipro bulb → Tuya LAN
Samsung AC → local DTLS / CoAP

←

Frigate · ctr / Intel iGPU
Camera ingest / recording
RTSP restream + MQTT → HA

Metrics · observability LXC
Exporters + Kubernetes vmagent → VictoriaMetrics
Grafana / HA node views; Beszel / Kuma monitoring

Local device protocols avoid cloud commands; a full WAN-outage acceptance test remains outstanding.

Solid arrows: application / data path → Dashed arrows: configuration / dependency ⇢ Not a live health map; see the inventory notes for verification limits.

Schemas are simplified; replicated storage and HA are not independent backups.